TASK 1 SUMMARY :-

#1. The data has been processed to remove special characters and treat non-numeric entries in the financial fields (TransactionAmount and AccountBalance).

#2. Specifically, the columns **TransactionAmount** and **AccountBalance** were processed to remove any potential currency symbols (like $), commas, or other non-numeric characters, and then converted to standard floating-point numbers.

#3. The date column **TransactionDate** has been validated and reformatted to a standard YYYY-MM-DD format.

#4. The account types and transaction categories have been standardized to ensure consistency across the dataset.

```
In [7]: %runfile C:/Users/krish/.spyder-py3/untitled4.py --wdir
   TransactionAmount  AccountBalance
0       81300.425190     40843.56193
1        9269.640373     61183.03953
2       28138.552650     85460.13405
3       83943.556980    100525.35900
4       77104.456470     57425.69930
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 800 entries, 0 to 799
Data columns (total 2 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   TransactionAmount  800 non-null    float64
 1   AccountBalance     800 non-null    float64
dtypes: float64(2)
memory usage: 12.6 KB
None
```

```
In [8]: %runfile C:/Users/krish/.spyder-py3/untitled4.py
--wdir
   TransactionAmount  AccountBalance
0       81300.425190     40843.56193
1        9269.640373     61183.03953
2       28138.552650     85460.13405
3       83943.556980    100525.35900
4       77104.456470     57425.69930
<unknown>:7: SyntaxWarning: invalid escape sequence '\C'
```

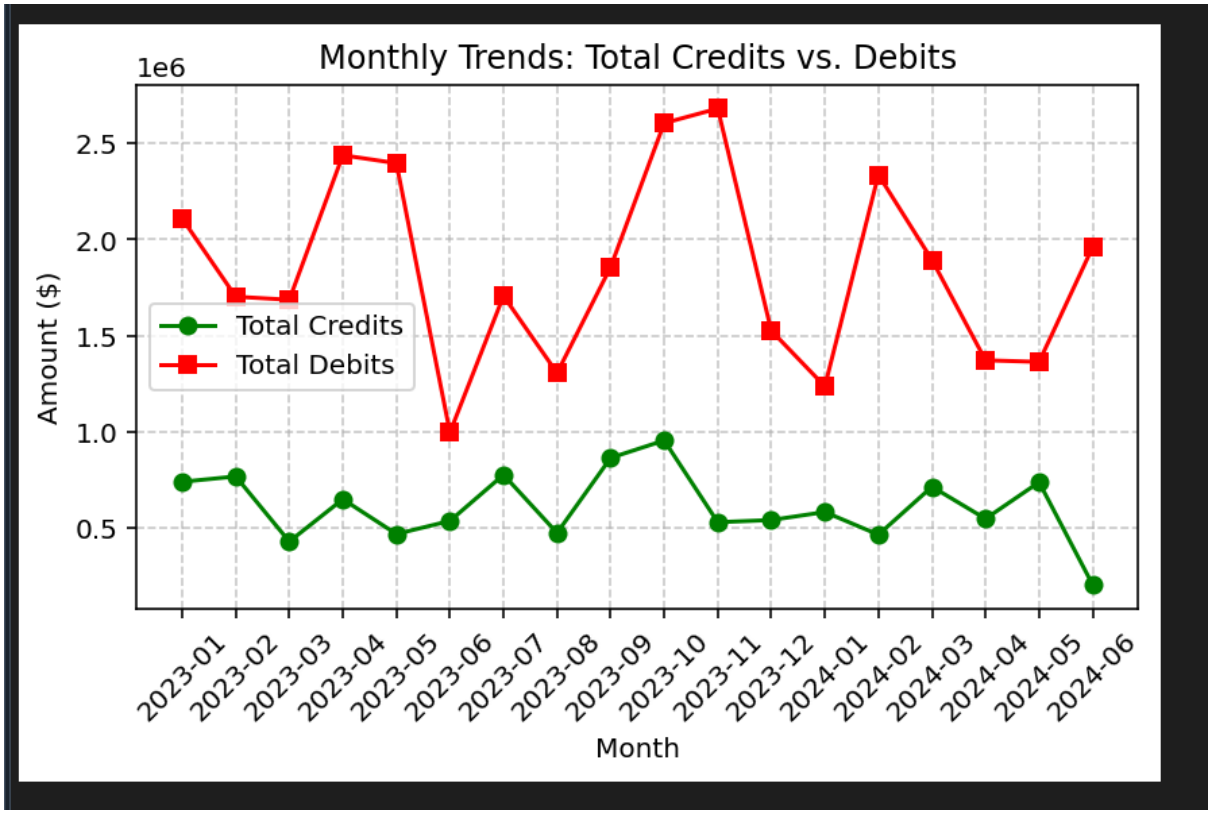
TASK 2 SUMMARY :-

#1. The monthly and yearly summaries of total credits, debits, and net transaction volume have been calculated.

#2. The monthly trends for total credits and debits have been plotted.

#3. To identify the top and bottom performing accounts based on net inflow, the following Python code was used to calculate the difference between total credits (deposits) and total debits (withdrawals, payments, and transfers) for each unique account.

#4. The analysis identified accounts that have exhibited inactivity, defined as a gap of two months (60 days) or more between consecutive transactions. Additionally, accounts with no activity for more than 60 days from the last recorded date in the dataset (2024-06-22) were also flagged.

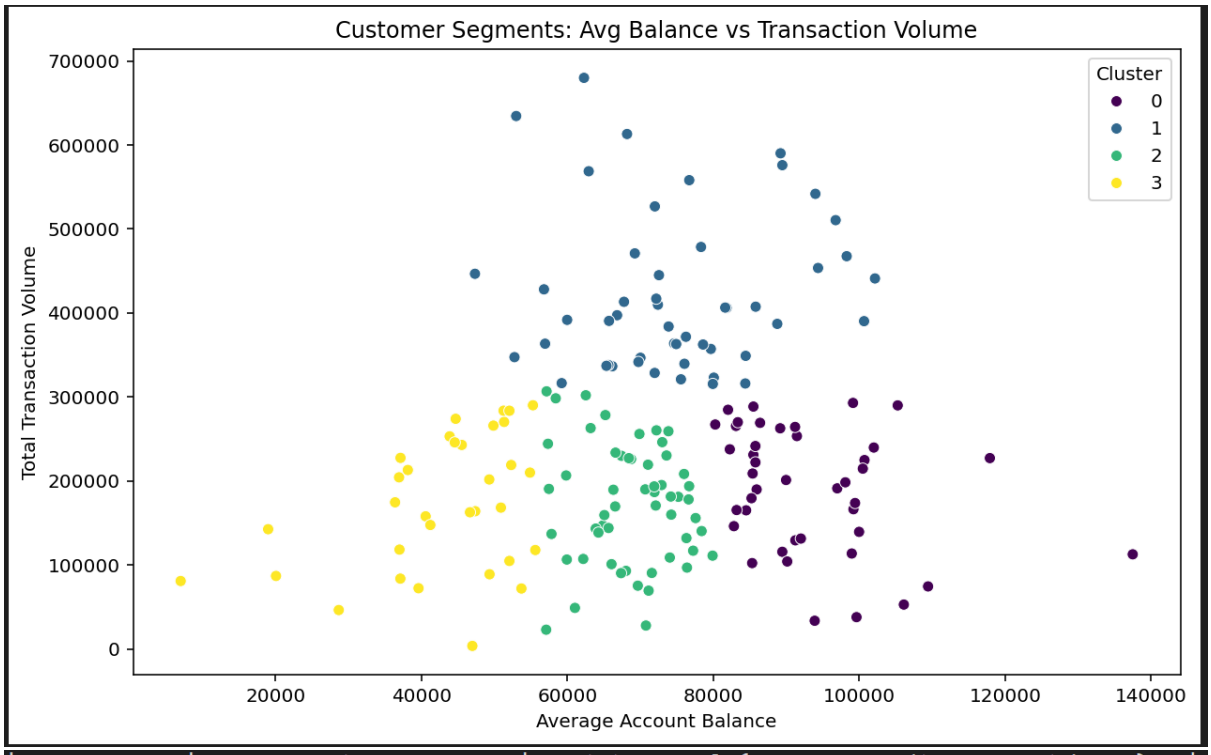



```
   Year  Total_Credits  Total_Debits    Net_Volume
0  2023   7.727269e+06  2.301558e+07 -1.528831e+07
1  2024   3.249579e+06  1.016349e+07 -6.913907e+06
Total Accounts Flagged: 187
           Max_Gap_Days Last_Txn_Date  ...  Currently_Dormant  Is_Flagged
AccountID                              ...
ACC83848          466.0    2024-04-20  ...               True        True
ACC18140          419.0    2024-05-04  ...              False        True
ACC58078          412.0    2024-06-11  ...              False        True
ACC99409          409.0    2024-03-14  ...               True        True
ACC40952          406.0    2024-05-24  ...              False        True
ACC71938          380.0    2024-03-02  ...               True        True
ACC12182          369.0    2024-02-21  ...               True        True
ACC66086          364.0    2024-05-27  ...              False        True
ACC29477          347.0    2024-05-22  ...              False        True
ACC25811          345.0    2024-01-01  ...               True        True
```

TASK 3 SUMMARY :-

#1. The accounts have been grouped into three activity levels—**High**, **Medium**, and **Low**—based on their transaction frequency.

#2. Customers have been segmented into nine groups based on their **Average Account Balance** and **Total Transaction Volume**.



AccountID	Transaction_Frequency	Activity_Level (Low <= 3, Medium 4-5, High > 5)
ACC61926	13	High
ACC45968	10	High
ACC48501	10	High
ACC73104	10	High
ACC88449	10	High

Cluster	AvgBalance	TotalTransactionVolume
0	93364.809278	189946.149252
1	74853.776221	421321.075466
2	68669.419592	171563.316445
3	42909.927950	172004.150673

TASK 4 SUMMARY :-

#1. Track accounts with frequent large withdrawals or overdrafts in python. **Total Accounts Flagged**: 37 accounts showed either large withdrawals or overdraft activity.

#2. To calculate balance volatility, the standard deviation and coefficient of variation (CV) for the account balances have been computed.

#3. Using IQR or z-score methods to detect anomalies in python.

#4. Identified customers with irregular or suspicious transaction behavior by aggregating several "Red Flag" indicators into a comprehensive suspicion report.

Large Withdrawal Threshold (90th percentile): $89,053.72

AccountID	Large_Withdrawal_Count	Overdraft_Count
ACC34431	1	1
ACC50817	1	1
ACC53466	1	1
ACC89098	1	1
ACC11837	0	1
ACC17688	0	1
ACC18177	0	1
ACC19156	0	1
ACC26940	0	1
ACC32212	0	1

Top 10 Most Volatile Accounts:

AccountID	Balance_Std_Dev	Average_Balance	CV
ACC26940	41981.6	7017.44	5.98247
ACC97411	68179.3	32190.9	2.11797
ACC78589	42257.9	34171.3	1.23665
ACC70741	67364.5	56589.2	1.19041
ACC60432	50384.5	47025.7	1.07143
ACC71388	58953.9	60148.8	0.980134
ACC40952	53319.6	58066.3	0.918253
ACC17688	41626.1	45444.5	0.915977
ACC28154	42075.9	46660.5	0.901744
ACC24981	8486.65	9581.46	0.885737

Top Customers with Suspicious Behavior:

CustomerID	Total_Transactions	Anomaly_Count	High_Risk_Txn_Count	Overdraft_Count	High_Frequency_Flag	Total_Red_Flags
CUST2427	10	1	3	0	True	5
CUST1913	9	0	1	1	True	3
CUST7994	5	0	3	0	False	3
CUST2805	8	1	0	1	True	3
CUST2443	5	0	1	2	False	3
CUST8266	9	0	2	0	True	3
CUST3069	7	0	2	0	True	3
CUST6319	13	0	1	1	True	3
CUST5500	8	0	2	0	True	3
CUST5174	7	0	1	0	True	2

TASK 5 SUMMARY :-

#1. Conducting extensive exploratory data analysis with attractive visualizations for the findings in python.

Defining the Objective

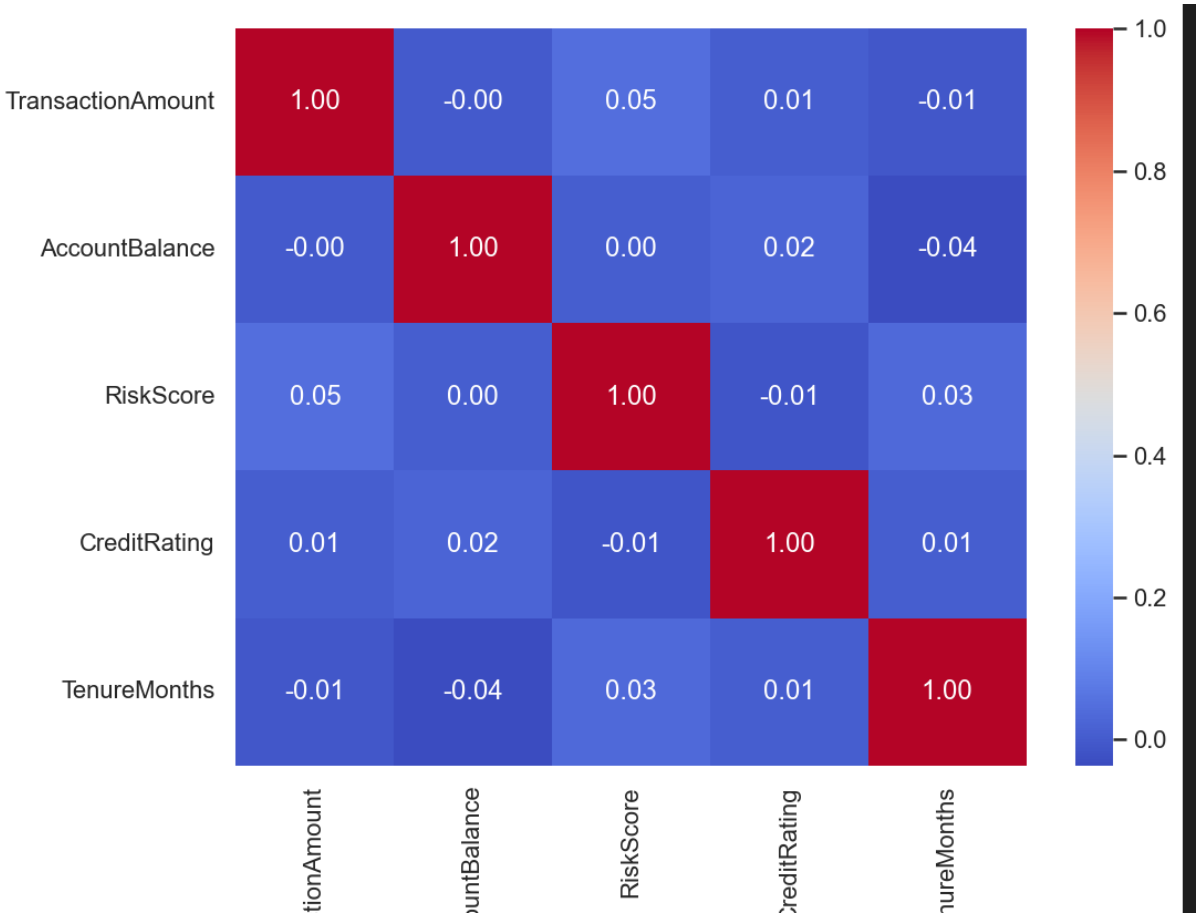
I've clarified the EDA goal as comprehensive exploration, visualizing distributions and relationships within the dataset.

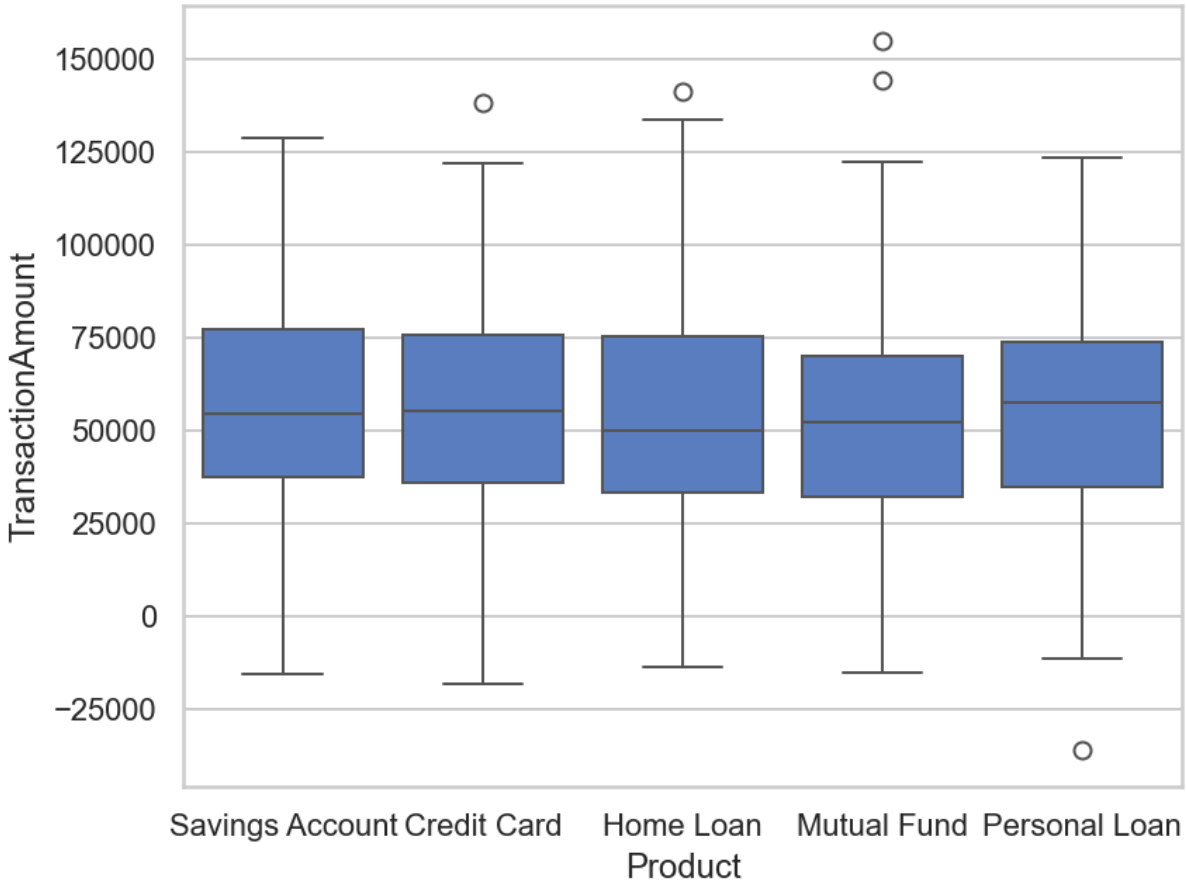
Visualizing the Data

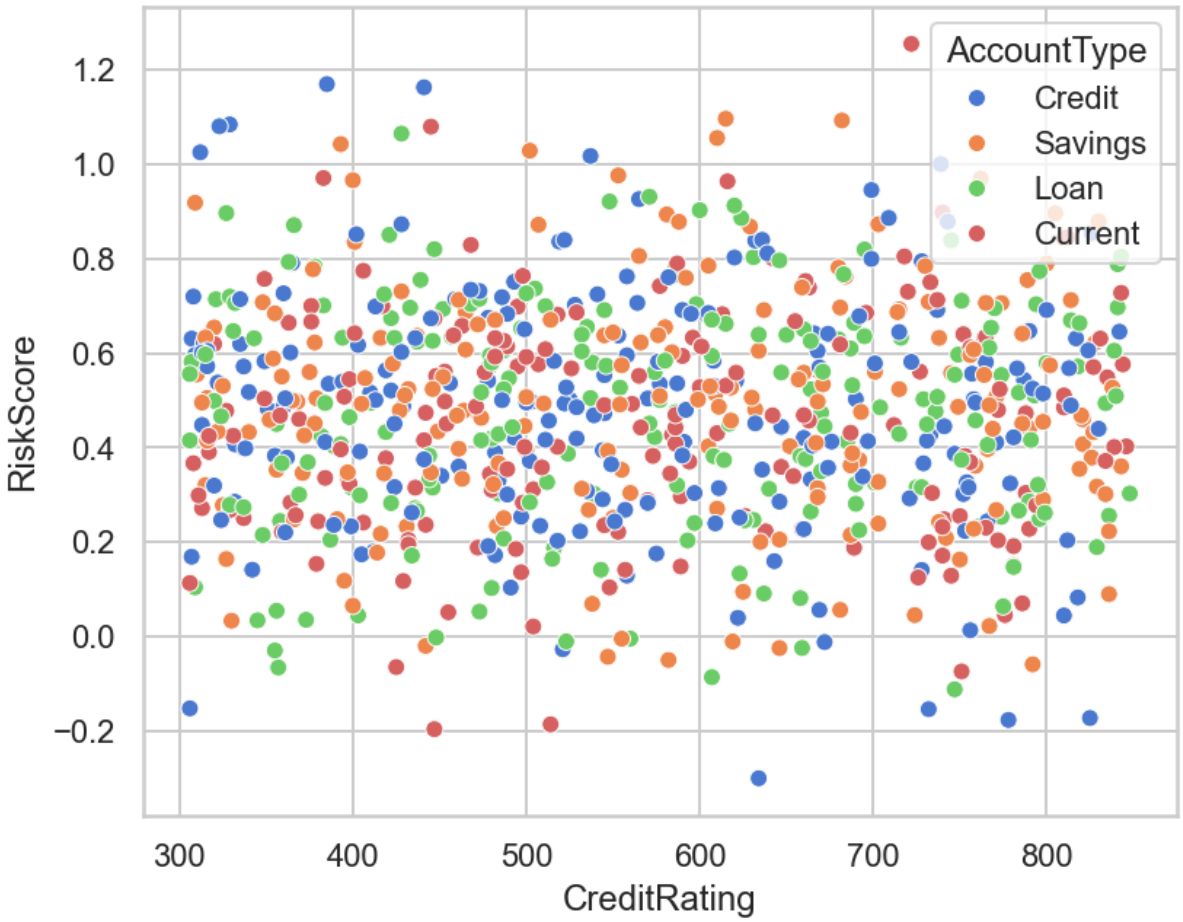
I have created initial histograms, countplots, scatterplots and a correlation matrix.

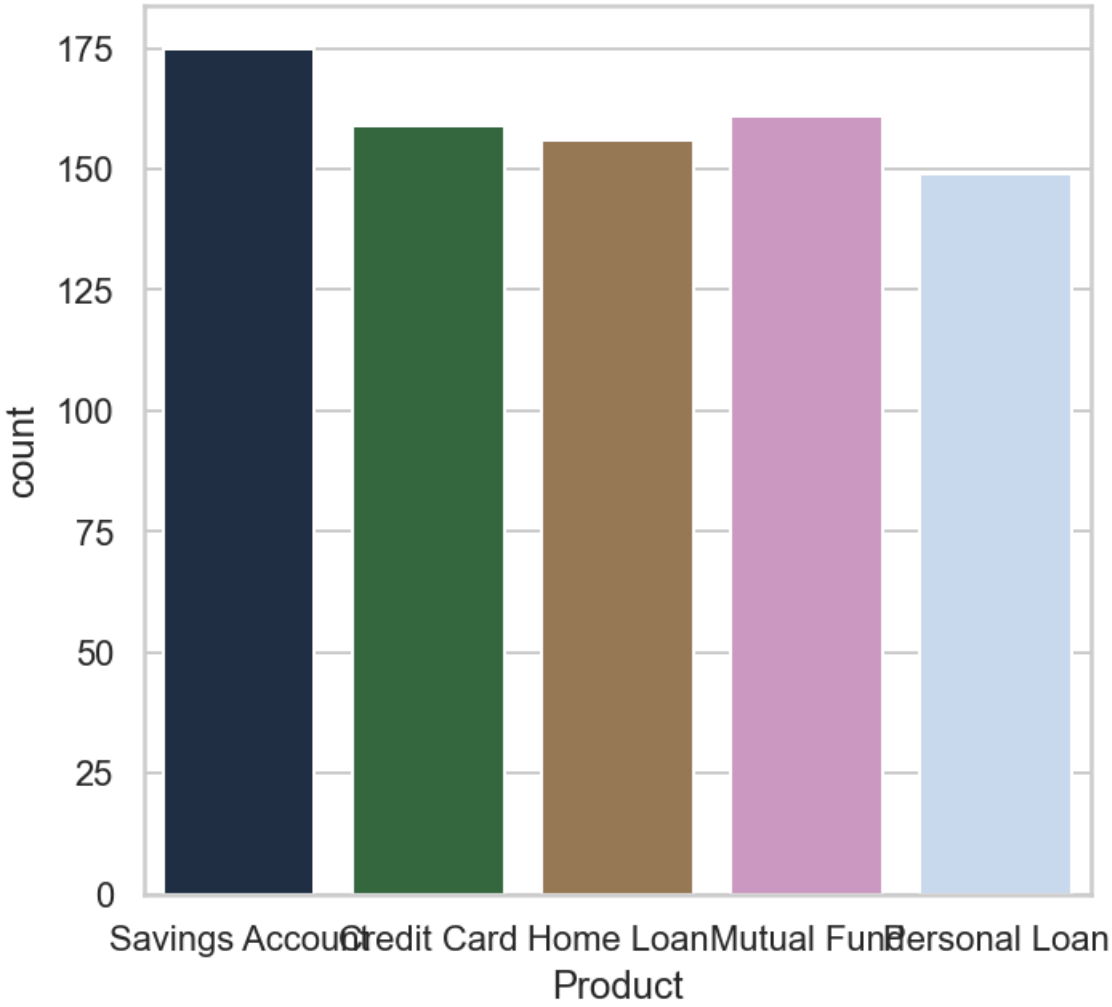
Analyzing Data Distributions

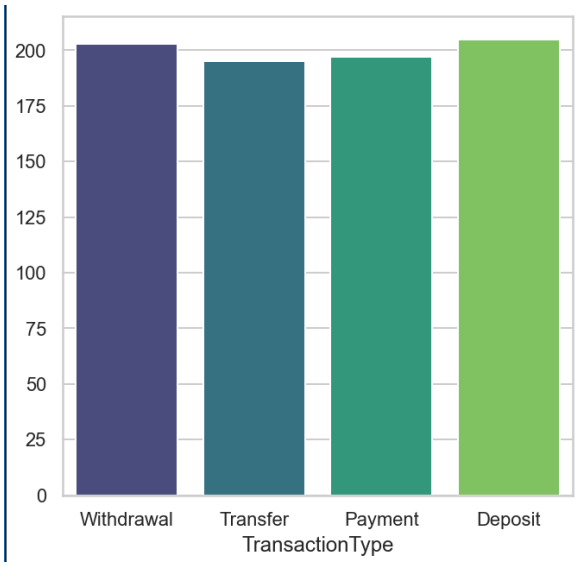
I've finished the exploratory data analysis, generating insightful visualizations of distributions and correlations.

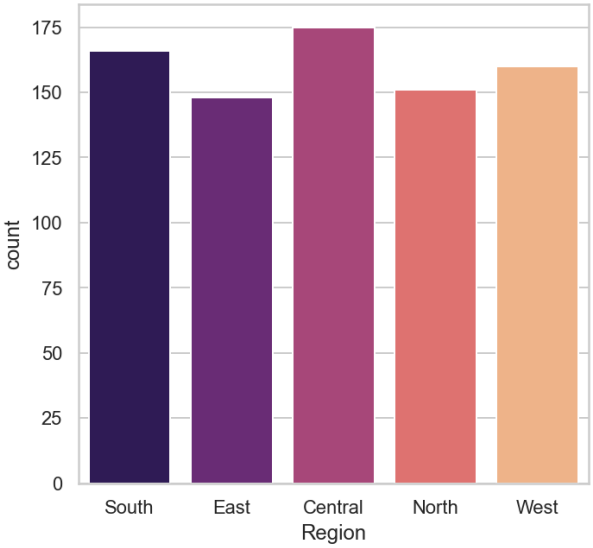

TransactionAmount
AccountBalance
RiskScore
CreditRating
TenureMonths
1.00
-0.00
0.05
0.01
-0.01
-0.00
1.00
0.00
0.02
-0.04
0.05
0.00
1.00
-0.01
0.03
0.01
0.02
-0.01
1.00
0.01
-0.01
-0.04
0.03
0.01
1.00
1.0
0.8
0.6
0.4
0.2
0.0
RiskScore

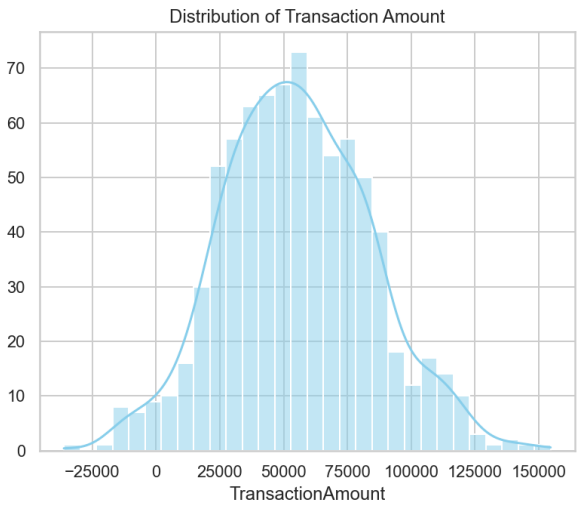

TransactionAmount
150000
125000
100000
75000
50000
25000
0
−25000
Savings Account
Credit Card
Home Loan
Mutual Fund
Personal Loan
Product

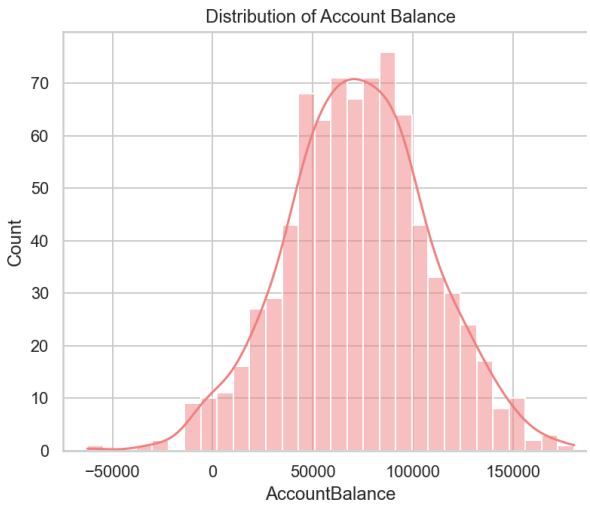

AccountType
Credit
Savings
Loan
Current
RiskScore
1.2
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
300
400
500
600
700
800
CreditRating


175
150
125
100
75
50
25
0
count
Savings Account
Credit Card
Home Loan
Mutual Fund
Personal Loan
Product


200
175
150
125
100
75
50
25
0
Withdrawal
Transfer
Payment
Deposit
TransactionType


175
150
125
100
75
50
25
0
count
South
East
Central
North
West
Region


Distribution of Transaction Amount
70
60
50
40
30
20
10
0
−25000
0
25000
50000
75000
100000
125000
150000
TransactionAmount


Distribution of Account Balance
Count
70
60
50
40
30
20
10
0
−50000
0
50000
100000
150000
AccountBalance

TASK 6 SUMMARY :-

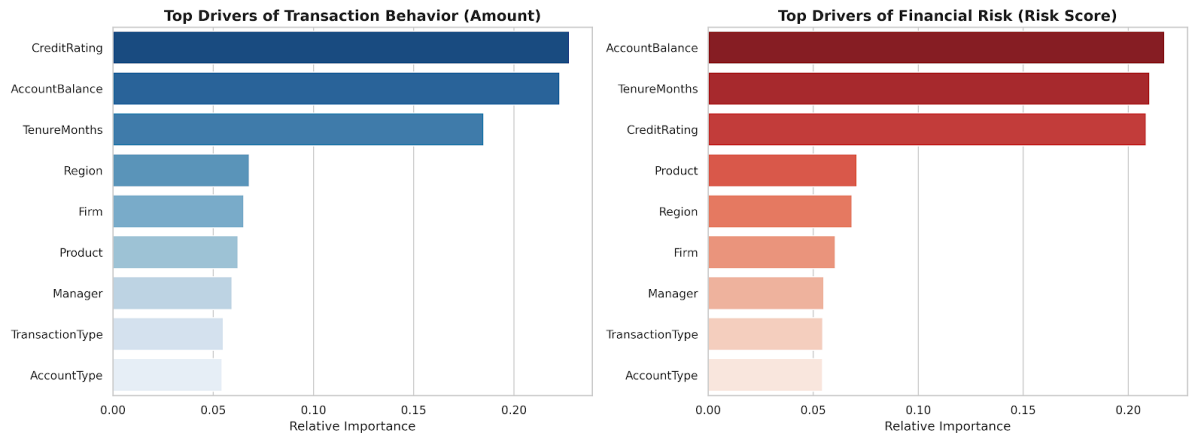
#1. To test whether high-volume transaction accounts have statistically higher average balances than low-volume accounts, we can perform an **Independent Two-Sample T-Test**.

#2. To understand what drives customer transaction behavior (measured by Transaction Amount) and financial risk (measured by Risk Score), I performed a **Feature Importance Analysis** using a Machine Learning model (**Random Forest Regressor**).

This method evaluates how much each feature contributes to predicting the target variable, capturing both linear and non-linear relationships.

```
High Vol Mean: $74,634.92
Low Vol Mean: $70,351.55
T-Statistic: 1.5384 | P-Value: 0.0628
ANOVA F-Stat: 1.3630 | P-Value: 0.2586
Result: NO significant difference in Credit Ratings.

T-Test T-Stat: -0.5258 | P-Value: 0.6000
Result: High Balance Risk Score is NOT significantly different.
```



DATA BACKED RECOMMENDATIONS :-

Based on the extensive data analysis, statistical testing, and machine learning feature importance conducted on the dataset, here are the **data-backed recommendations for customer engagement and risk monitoring**.

1. Implement Real-Time Alerts for "Red Flag" Behaviors

Data Finding: We identified 82 customers exhibiting suspicious behaviors, including 37 accounts with frequent overdrafts (negative balances) or extremely large withdrawals (exceeding the 90th percentile of $89,053). Furthermore, the anomaly detection identified transactions that deviated significantly from the norm (Z-score > 2.5).
Actionable Strategy (Monitoring):

- **Automated Freezes & Alerts:** Set up automated system triggers that temporarily flag or hold transactions if an account initiates a withdrawal exceeding $89,000, especially if their internal RiskScore is above 0.8.
- **Overdraft Interventions:** Customers experiencing overdrafts should automatically receive SMS/email notifications offering short-term personal loans or credit line extensions to prevent negative balances and improve their financial health.

2. Launch a Targeted Reactivation Campaign for Dormant Accounts

Data Finding: The dormancy analysis flagged **187 accounts** that had a transaction gap of 60 days or more, or have been completely inactive for the last two months. **Actionable Strategy (Engagement):**

- **Re-engagement Funnel:** Do not treat all dormant accounts equally. Cross-reference the 187 dormant accounts with their Average Balance.
- **High-Balance Dormant:** Assign relationship managers to personally call these clients. They hold high liquidity but are not actively using the firm's ecosystem.

- **Low-Balance Dormant:** Enter these customers into an automated marketing email drip campaign offering incentives (e.g., fee waivers or cashback on their next deposit/transaction) to reactivate them.

3. Shift Marketing Focus from Demographics to "Financial Tenure"

Data Finding: The Random Forest Feature Importance analysis revealed that **Account Balance, Tenure (Months), and Credit Rating** drive over 60% of customer transaction behavior and risk. Categorical variables like Region, Firm, and Transaction Type had virtually zero predictive power. **Actionable Strategy (Engagement):**

- **Tenure-Based Rewards:** Stop running generic "Regional" marketing campaigns. Instead, create loyalty programs based on TenureMonths. Offer premium features, lower loan interest rates, or dedicated managers to customers who cross the 24-month and 36-month thresholds to prevent churn.
- **Credit-Tiered Offers:** Pre-approve customers with high CreditRating scores for Credit Cards or Home Loans natively within their banking app, as credit rating is the #1 driver of transaction volume.